SHAHINA

Experimental Nuclear Astrophysicist
Postdoctoral Research Associate, Plasma Science and Fusion Center, Massachusetts Institute of Technology
shahina@mit.edu — +1(574)386-5699 — Cambridge, MA 02139

ACADEMIC APPOINTMENTS

Massachusetts Institute of Technology, Plasma Science and Fusion Center Jan. 2025–Present
Postdoctoral Research Associate
High-energy-density plasma physics, fusion diagnostics, and nuclear cross-section measurements.

University of Notre Dame, Department of Physics and Astronomy Aug. 2018–Apr. 2024
Graduate Research Assistant

EDUCATION

University of Notre Dame, Notre Dame, IN 2018–2024
Ph.D., Physics and Astronomy
Dissertation: *Stellar neutron sources for the s-process nucleosynthesis*
Advisor: Prof. Michael Wiescher

University of Notre Dame, Notre Dame, IN 2018–2020
M.S., Physics and Astronomy

Indian Institute of Science Education and Research (IISER) Mohali, India 2017–2018
M.S., Physics
Thesis: *Air-shower Multi-Mesh Proportional Counter*
Advisor: Prof. Satyajit Jena

Indian Institute of Science Education and Research (IISER) Mohali, India 2013–2017
B.S., Physics

RESEARCH INTERESTS

My research interests lie in low-energy nuclear physics, high-energy-density plasma physics, detector development, and computational analysis to measure nuclear reactions that govern stellar evolution and nucleosynthesis.

- **Experimental nuclear astrophysics:** Direct measurements of radiative-capture and (α, n) reactions relevant to the slow and rapid neutron capture process, the Carbon-Nitrogen-Oxygen (CNO) cycle in Sun, and stellar evolution.
- **High-energy-density plasma nuclear science:** Development of plasma platforms and diagnostics for nuclear-reaction measurements at OMEGA, the Z machine, and the National Ignition Facility.
- **Detector development and neutron spectroscopy:** Neutron, gamma-ray, charged-particle, and CR-39 based detection methods for accelerator-based cross-section measurements.
- **Reaction modeling and data analysis:** R-matrix and Hauser-Feshbach analyses, MESA stellar-evolution code, Monte Carlo particle transport, and machine-learning methods for nuclear data analysis and neutron-spectrum unfolding.

CORE EXPERTISE

Neutron spectroscopy and unfolding (MLEM, Bayesian inference) • **Gamma-ray and charged-particle detection** (HECTOR, CR-39, scintillators) • **Accelerator-based nuclear experiments** (CASPAR, LEIA) • **High-energy-density plasma diagnostics** (OMEGA, NIF, Z) • **Monte Carlo transport and nuclear data** (MCNP/OpenMC, Geant4, ENDF).

PUBLICATIONS

Manuscripts submitted and in preparation

1. *Advancing the MIT Linear Electrostatic Ion Accelerator (LEIA) used for developing diagnostics for OMEGA, Z and NIF for fusion cross-section measurements*
   Shahina, T. Evans, K. Bauer, B.I. Buschmann, M. Cufari, S. Dannhoff, A. DeVault, E. Edwards, B. Foo, N. Keer, Y. Lawrence, C. Shuldberg, N. Vanderloo, J. Vargas, C. Wink, J.A. Frenje, and M. Gatu Johnson.
   Submitted to *Review of Scientific Instruments*.

2. *The $^{26}Mg(\alpha, n)^{29}Si$ reaction at low energies*
   U. Bajpai, **Shahina**, K.T. Macon, A. Boeltzig, R.J. deBoer, M. Febbraro, S. Aguilar, T. Anderson, S.P. Burcher, Y. Chen, A. Clark, A. Dombos, C. Dulal, B. Frentz, G. Gilardy, O. Gomez, J. Görres, A. Gula, S. Henderson, K. Howard, J.M. Kovoor, K.L. Jones, J. Kelley, R. Kelmar, Q. Liu, A. Majumdar, K. Manukyan, L. Morales, S. Mosby, S. Moylan, A. Nelson, S. Pain, P. Scholz, C. Reingold, M. Renaud, N. Sensharma, C. Seymour, E. Stech, W. Tan, R. Toomey, B. Van de Kolk, J. Wilkinson, and M. Wiescher.
   Manuscript in preparation (2026).

First-author refereed publications

1. *Low energy measurement of the $^{25}Mg(\alpha, n)^{28}Si$ reaction via neutron spectroscopy*
   Shahina, A. Boeltzig, K.T. Macon, R.J. deBoer, M. Febbraro, S. Aguilar, T. Anderson, S.P. Burcher, Y. Chen, A. Clark, A. Dombos, C. Dulal, B. Frentz, G. Gilardy, O. Gomez, J. Görres, A. Gula, S. Henderson, K. Howard, J.M. Kovoor, K.L. Jones, J. Kelley, R. Kelmar, Q. Liu, A. Majumdar, K. Manukyan, L. Morales, S. Mosby, S. Moylan, A. Nelson, S. Pain, C. Reingold, M. Renaud, N. Sensharma, C. Seymour, R. Toomey, B. Van de Kolk, J. Wilkinson, and M. Wiescher.
   Physical Review C **113**, 035802 (March 2026).
2. *Strength measurement of the 830 keV resonance in the $^{22}Ne(\alpha, n)^{25}Mg$ reaction using a stilbene detector*
   Shahina, R.J. deBoer, J. Görres, R. Fang, M. Febbraro, R. Kelmar, M. Matney, K. Manukyan, J.T. Nattress, E. Stech, and M. Wiescher.
   Physical Review C **110**, 015801 (July 2024).
3. *Direct measurement of the low-energy resonances in the $^{22}Ne(\alpha, \gamma)^{26}Mg$ reaction*
   Shahina, J. Görres, D. Robertson, M. Couder, O. Gomez, A. Gula, M. Hanhardt, T. Kadlecek, R. Kelmar, P. Scholz, A. Simon, E. Stech, F. Strieder, and M. Wiescher.
   Physical Review C **106**, 025805 (August 2022).

Co-authored refereed publications

1. *Comprehensive study of d+^{6}Li via observing correlated outgoing neutrons, γ-rays and charged particles*
   S.N. Paneru, H.Y. Lee, C. Prokop, S.A. Kuvin, C. Fichtl, P. Gastis, G.M. Hale, E. Leal-Cidoncha, M. Mosby, M. Paris, M. Febbraro, T.T. King, J. Nattress, T.J. Ruland, R.J. deBoer, E. Stech, T. Bailey, C. Boomershine, S. Carmichael, A. Clark, R. Fang, J. Görres, R. Kelmar, K. Lee, K. Manukyan, M. Matney, J. McDonaugh, A. Miller, A. Nelson, P. O'Malley, D. Robertson, **Shahina**, W. Tan, W. von Seeger, and A. Roberts.
   Physical Review C **110**, 044603 (October 2024).
2. *Coincident measurement of the ^{12}C+^{12}C fusion cross section via the differential thick target technique*
   W.P. Tan, O. Gomez, A. Gula, K. Lee, A. Majumdar, S. Moylan, **Shahina**, and M. Wiescher.
   Physical Review C **110**, 035808 (September 2024).
3. *Measurement of the $^{13}C(\alpha, n)^{16}O$ differential cross section from 0.8 to 6.5 MeV*
   R.J. deBoer, M. Febbraro, D. Bardayan, C. Boomershine, K. Brandenburg, C. Brune, S. Coil, J. Derkin, S. Dede, F. Fang, A. Fritsch, A. Gula, Gy. Gyürky, B. Hackett, G. Hamad, Y. Jones-Alberty, R. Kelmar, K. Manukyan, M. Matney, J. McDonaugh, Z. Meisel, S. Moylan, J. Nattress, D. Odell, P. O'Malley, D. Robertson, **Shahina**, N. Singh, K. Smith, M. Smith, E. Stech, W. Tan, and M. Wiescher.
   Physical Review Letters **132**, 062702 (February 2024).
4. *^{10}B+α reactions at low energies*
   A. Gula, R.J. deBoer, S. Aguilar, J. Arroyo, C. Boomershine, B. Frentz, J. Görres, S. Henderson, R. Kelmar, S. McGuinness, K.V. Manukyan, S. Moylan, D. Robertson, C. Seymour, **Shahina**, E. Stech, W. Tan, J. Wilkinson, and M. Wiescher.
   Physical Review C **107**, 025805 (February 2023).
5. *First near-threshold measurements of the $^{13}C(\alpha, n)^{16}O$ reaction for low-background-environment characterization*
   R.J. deBoer, A. Gula, M. Febbraro, K. Brandenburg, C.R. Brune, J. Görres, Gy. Györky, R. Kelmar, K. Manukyan, Z. Meisel, D. Odell, M.T. Pigni, **Shahina**, E. Stech, W. Tan, and M. Wiescher.
   Physical Review C **106**, 055808 (November 2022).
6. *Performance of neutron spectrum unfolding using deuterated liquid scintillator*
   M. Febbraro, B. Becker, R.J. deBoer, K. Brandenburg, C. Brune, K.A. Chipps, T. Danley, A. Di Fulvio, Y. Jones-Alberty, K.T. Macon, Z. Meisel, T.N. Massey, R.J. Newby, S.D. Pain, S. Paneru, **Shahina**, M.S. Smith, D. Soltesz, S.K. Subedi, I. Sultana, and R. Toomey.
   Nuclear Instruments and Methods in Physics Research A **989**, 164824 (February 2022).

7. *Light response of poly(ethylene 2,6-naphthalate) to neutrons*
B. Hackett, R. deBoer, Y. Efremenko, M. Febbraro, J. Nattress, D. Bardayan, C. Boomershine, K. Brandenburg, S. Dede, J. Derkin, R. Fang, A. Fritsch, A. Gula, Gy. Györky, G. Hamad, Y. Jones-Alberty, R. Kelmar, K. Manukyan, M. Matney, J. McDonaugh, S. Moylan, P. O'Malley, **Shahina**, and N. Singh.
Nuclear Instruments and Methods in Physics Research A 1069 (2024) 169900

8. *Determination of hexadecapole β_4 deformation of the light-mass nucleus ^{24}Mg using quasi-elastic measurement*
Y.K. Gupta, B.K. Nayak, U. Garg, N. Sensharma, **Shahina**, R. Gandhi, D.C. Biswas, M. Senyigit, K.B. Howard, W. Tan, P.D. O'Malley, K. Hagino, M. Smith, O. Hall, M. Hall, R.J. deBoer, K. Ostdiek, Q. Liu, A. Long, J. Hu, T. Anderson, M. Skulski, W. Lu, E. Lamere, S. Lyons, B. Frentz, A. Gyurjinyan, B. Van de Kolk, and C. Seymour.
Proceedings of the DAE Symposium on Nuclear Physics **58** (2018).

RESEARCH EXPERIENCE

Massachusetts Institute of Technology, Cambridge, MA Jan. 2025–Present
Postdoctoral Research Associate, Plasma Science and Fusion Center

High-energy-density nuclear physics and accelerator diagnostics

- Developing a high-energy-density plasma platform for measurements of nuclear reactions relevant to the CNO cycle.
- Advancing the MIT Linear Electrostatic Ion Accelerator (LEIA) for the development and calibration of diagnostics used in fusion cross-section measurements at OMEGA, the Z machine, and the National Ignition Facility.
- Cross-section measurement of the CNO surrogate ^{13}C+d reaction at the MIT LEIA accelerator.

University of Notre Dame, Notre Dame, IN Aug. 2018–Apr. 2024
Graduate Research Assistant, Department of Physics and Astronomy

Direct measurement of low-energy resonances in the ^{22}Ne$(\alpha,\gamma)^{26}$Mg reaction

- Measured the 650 and 830 keV resonances at the Compact Accelerator System for Performing Astrophysical Research (CASPAR), located 4,850 ft underground at the Sanford Underground Research Facility.
- Used the HECTOR gamma-summing detector to constrain stellar neutron-source reaction rates.

Low-energy measurements of the ^{25,26}Mg$(\alpha,n)^{28,29}$Si reactions

- Performed precision cross-section measurements at the University of Notre Dame Nuclear Science Laboratory.
- Applied neutron spectroscopy with the ORNL Deuterated Spectroscopic Array (ODeSA) and maximum-likelihood expectation-maximization methods for neutron-spectrum unfolding.

Measurement of the 830 keV resonance in the ^{22}Ne$(\alpha,n)^{25}$Mg reaction

- Measured the resonance strength using a stilbene scintillator detector, contributing to improved constraints on neutron production for s-process nucleosynthesis.

Indian Institute of Science Education and Research (IISER) Mohali, India 2017–2018
M.S. Researcher, Advisor: Prof. Satyajit Jena

- Simulated and designed an air-shower Multi-Mesh Proportional Counter for vetoing external particles in cosmic radiation.

Bhabha Atomic Research Centre, Mumbai, India Summer 2017
Experimental Nuclear Physics Researcher, Advisor: Dr. Y.K. Gupta

- Analyzed experimentally extracted barrier distributions for heavy-ion fusion and quantified entrance-channel effects from target and projectile structure.

Inter-University Accelerator Centre, New Delhi, India Summer 2016
Experimental Nuclear Physics Researcher, Advisor: Dr. Jagdish Gehlot

- Assembled and characterized a Multi-Wire Proportional Counter using an ^{241}Am alpha source for position calibration.

Centre for Theoretical Physics, Jamia Millia Islamia, New Delhi, India Summer 2015
Classical Mechanics Researcher, Advisor: Prof. Ratin Adhikari

- Studied inverse variational problems, including construction of Lagrangians from equations of motion and transformations of generalized coordinates.

RESEARCH PROPOSALS AND FACILITY ACCESS

Co-Principal Investigator, NLUF Shot Proposal Awarded 2025
Nuclear astrophysics experiments using high energy density plasmas
Research Topic Area: Plasma and Nuclear Physics.

Co-Principal Investigator, NIF Discovery Science Proposal Awarded 2025
Exploration of Stellar Nucleosynthesis and Basic Nuclear Science using High Energy Density Plasmas at the NIF

RESEARCH MENTORING AND ADVISING

Utkarsh Bajpai, Graduate Student, University of Notre Dame Current
Co-mentor with Prof. Michael Wiescher on data analysis for the ^{26}Mg$(\alpha, n)^{29}$Si reaction.

Andrew Lanzrath, Graduate Student, Massachusetts Institute of Technology Current
Advise on the development and analysis of nuclear astrophysics experiments using high-energy-density plasma platforms at OMEGA and NIF.

Undergraduate Students Current
Sayak Moulic (IIT Kharagpur), Krish Majhi (IISER Bhopal), and Ayushman Joshi (IISER Pune). Advising them on the CR-39 detector data analysis from the MIT accelerator measurement of the ^{13}C+d cross section.

TEACHING EXPERIENCE

Teaching areas: Introductory and calculus-based physics, nuclear physics, nuclear astrophysics, radiation detection, and computational/data-analysis methods.

University of Notre Dame, Notre Dame, IN

Lead Instructor, Physics Teaching Practicum Spring 2023

- Delivered lectures for Physics for Life Sciences under faculty observation and incorporated feedback on course organization, pacing, and student engagement.

Lead Laboratory Teaching Assistant, Physics for Life Sciences Laboratory Fall 2018–Spring 2020

- Led laboratory instruction, demonstrated experiments and data analysis, prepared equipment, and troubleshot experimental setups for undergraduate students.

Teaching Assistant, Engineering Physics I Fall 2019

- Led two weekly tutorial sessions for approximately 100 undergraduate students, combining review, collaborative problem-solving, homework guidance, and individual office-hour mentoring.

NROTC Tutor, Physics, Calculus, and Chemistry Spring 2022–Present

- Tutor Notre Dame Naval Reserve Officer Training Corps students in introductory physics, calculus, and chemistry through twice-weekly individual and small-group sessions.

Physics Tutor Spring 2022–Present

- Provide one-on-one support in undergraduate physics courses, emphasizing conceptual understanding and structured problem-solving.

INVITED TALKS AND CONFERENCE PRESENTATIONS

Invited talks

- "Developing a high-energy-density plasma platform for studying nuclear reactions relevant to the CNO cycle," 67th Annual Meeting of the APS Division of Plasma Physics, Long Beach, CA, Nov. 17–21, 2025.
- "Stellar neutron sources for the s-process nucleosynthesis," Cyclotron Institute, Texas A&M University, College Station, TX, Jan. 23, 2024.
- "Stellar neutron sources for the s-process nucleosynthesis," School-cum-Workshop on Low Energy Nuclear Astrophysics, Saha Institute of Nuclear Physics, Kolkata, India, Nov. 7–10, 2023.
- "Stellar neutron sources for the s-process nucleosynthesis," Nuclear Physics Seminar, University of Notre Dame, Notre Dame, IN, Sept. 11, 2023.

- "Measurement of the ^{25}Mg$(\alpha, n)^{28}$Si reaction cross section at low energy," R-Matrix Workshop on Methods and Applications, June 21–24, 2021.

Contributed talks

- "Measurement of the low-energy resonances in the ^{22}Ne$(\alpha, \gamma)^{26}$Mg and ^{22}Ne$(\alpha, n)^{25}$Mg reactions," International Symposium on Nuclear Astrophysics, Manipal, India, Oct. 30–Nov. 3, 2023.
- "Direct measurement of the low-energy resonances in the ^{22}Ne$(\alpha, \gamma)^{26}$Mg reaction," Nuclei in the Cosmos, Institute for Basic Science, Daejeon, South Korea, Sept. 17–22, 2023.
- "Measurement of the low-energy resonances in the ^{22}Ne$(\alpha, \gamma)^{26}$Mg and ^{22}Ne$(\alpha, n)^{25}$Mg reactions," Fall Meeting of the APS Division of Nuclear Physics, New Orleans, LA, Oct. 27–30, 2022.
- "Direct measurement of the low-energy resonances in the ^{22}Ne$(\alpha, \gamma)^{26}$Mg reaction," Nuclear Physics in Astrophysics X (NPA-X), CERN, Geneva, Switzerland, Sept. 5–9, 2022.
- "Direct measurement of the low-energy resonances in the ^{22}Ne$(\alpha, \gamma)^{26}$Mg reaction," 11th European Summer School on Experimental Nuclear Astrophysics, Catania, Italy, June 12–19, 2022.
- "Direct measurement of the low-energy resonances in the ^{22}Ne$(\alpha, \gamma)^{26}$Mg reaction," JINA-CEE Frontiers in Nuclear Astrophysics Meeting, University of Notre Dame, Notre Dame, IN, May 25–27, 2022.
- "Direct measurement of the low-energy resonances in the ^{22}Ne$(\alpha, \gamma)^{26}$Mg reaction," Fall Meeting of the APS Division of Nuclear Physics, Oct. 11–14, 2021.
- "Measurement of the ^{25}Mg$(\alpha, n)^{28}$Si reaction cross section at low energy," Fall Meeting of the APS Division of Nuclear Physics, Oct. 29–Nov. 1, 2020.
- "Measurement of the ^{25}Mg$(\alpha, n)^{28}$Si reaction cross section at low energy," Nuclear Physics Seminar, University of Notre Dame, Notre Dame, IN, Aug. 24, 2020.

Poster presentations

- **Shahina** et al., "Advancing the MIT Linear Electrostatic Ion Accelerator (LEIA) used for developing diagnostics for OMEGA, Z and NIF for fusion cross-section measurements," 26th Topical Conference on High Temperature Plasma Diagnostics, Cambridge, MA, June 7–11, 2026.
- **Shahina** et al., "Developing a high-energy-density plasma platform for studying nuclear reactions relevant to the CNO cycle," Omega Laser Facility Users Group 2026 Workshop, Rochester, NY, Apr. 28–30, 2026.
- **Shahina** et al., "Developing a high-energy-density plasma platform for studying nuclear reactions relevant to the CNO cycle," National Ignition Facility and Jupiter Laser Facility User Groups Meeting, Livermore, CA, Feb. 10–12, 2026.
- **Shahina** et al., "Developing the high-energy-density plasma platform for studying nuclear reactions relevant to the CNO cycle," Omega Laser Facility Users Group 2025 Workshop, Rochester, NY, May 20–22, 2025.

HONORS AND AWARDS

Best Poster Award, Postdoctoral Category, High-Energy-Density Science Summer School 2025
San Diego, CA

Innovation in Science Pursuit for Inspired Research (INSPIRE) Fellowship 2013–2018
Nationally competitive five-year fellowship awarded by the Department of Science and Technology, India.

Selected IISER Mohali Representative, Young Scientists' Conference 2015
India International Science Festival, Indian Institute of Technology Delhi, New Delhi, India.

Joint Entrance Examination (JEE) Mains and Advanced 2013
Ranked among the top 1% nationally in India's engineering entrance examinations.

ACADEMIC SERVICE, LEADERSHIP, AND OUTREACH

Upward Bound, University of Notre Dame Fall 2023
Tutor

- Mentored high-school students in academic planning, study skills, SAT/ACT preparation, and college applications.

JINA-CEE Frontiers in Nuclear Astrophysics Conference May 25–27, 2022
Organizing Committee Member, University of Notre Dame

Indiana Science Communication Day, Indiana Statehouse Feb. 24, 2022
Science Policy Initiative Presenter

- Presented Ph.D. research to legislators and public audiences, including the state senator representing South Bend.

Graduate Student Government, University of Notre Dame 2021–2022
Sustainability Chair

- Managed graduate-student garden plots, strengthened composting practices, and expanded campus recycling initiatives.

PROFESSIONAL AFFILIATIONS

- American Physical Society, Member 2018–Present
- Joint Institute for Nuclear Astrophysics–Center for the Evolution of the Elements (JINA-CEE), Member 2018–Present
- International Research Network for Nuclear Astrophysics (IReNA), Member 2021–Present
- CASPAR Collaboration, Member 2018–Present
- Association for Women in Science, Member 2018–Present
- Science Policy Initiative at Notre Dame, Member 2018–Present

SUMMER SCHOOLS AND WORKSHOPS

- High-Energy-Density Science Summer School, San Diego, CA, 2025.
- National Nuclear Physics Summer School, Massachusetts Institute of Technology, Cambridge, MA, July 11–22, 2022.
- 11th European Summer School on Experimental Nuclear Astrophysics, Catania, Italy, June 12–19, 2022.
- 19th Exotic Beam Summer School, University of Notre Dame, Notre Dame, IN, June 6–10, 2022.
- R-Matrix Workshop on Methods and Applications, June 21–24, 2021.
- Nuclear TALENT Course: Machine Learning and Data Analysis for Nuclear Physics, ECT*, Trento, Italy, June 22–July 3, 2020.
- AI for Nuclear Physics Workshop, Thomas Jefferson National Accelerator Facility, Newport News, VA, Mar. 4–6, 2020.

TECHNICAL EXPERTISE

Experimental methods: Accelerator-based nuclear-reaction cross-section measurements; gamma-ray, neutron, charged-particle, and CR-39 detector characterization and nuclear spectroscopy.

Modeling and simulation: R-matrix and Hauser-Feshbach calculations; MESA stellar-evolution code; GEANT4, Garfield, MCNP, OpenMC, and SRIM.

Programming and data analysis: C, C++, Python, Fortran, ROOT, MATLAB, PyTorch, LabVIEW, machine learning, and neural networks.

Design and nuclear data: SolidWorks, LaTeX, ENDF, EXFOR, and NNDC.

REFERENCES

Dr. Johan Frenje
Senior Research Scientist and HEDP Division Head, MIT Plasma Science and Fusion Center
jfrenje@psfc.mit.edu

Dr. Maria Gatu Johnson
Principal Research Scientist and Assistant Director, MIT Plasma Science and Fusion Center
gatu@psfc.mit.edu

Prof. Michael Wiescher
Department of Physics and Astronomy, University of Notre Dame
Michael.C.Wiescher.1@nd.edu

Dr. Michael T. Febbraro
Air Force Institute of Technology, Wright-Patterson Air Force Base
michael.febbraro@us.af.mil

Dr. Richard J. deBoer
Department of Physics and Astronomy, University of Notre Dame
Richard.J.deBoer.12@nd.edu